Abdullah Khaled

Embedded systems engineer with robotics, vision, and on-device ML experience

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Summary

- Embedded systems engineer with hands-on experience building robotics automation, sensor fusion, and on-device inference systems.
- Delivered reliability engineering for competition-grade robotics, led distributed software teams, and developed low-cost vision hardware for real-world monitoring applications.

Skills

Hardware & Prototyping: Fusion 360, Onshape, NI Multisim, KiCad, EasyEDA, soldering, PCB prototyping

Software & Machine Learning: ROS, Python, Java, JavaScript, TensorFlow, OpenCV

Systems & Methods: Kalman filtering, finite state machines, mechanism simulation, control theory, prompt engineering

Work Experience

Math and Reading Instructor, Gideon Math and Reading Center – Frisco, TX — June 2024 – present

- Designed individualized instruction plans using student performance data to identify learning gaps and accelerate progress.
- Managed daily small-group sessions, adapting delivery and materials across multiple grade levels.
- Strengthened progress tracking by implementing structured checkpoints and targeted review.

Projects

OralVision – **Conrad Challenge; Diamond Challenge** — Oct 2025 – present

- Developed a privacy-preserving oral disease detection prototype using federated learning for centralized model training and on-device inference.
- Designed and programmed a low-cost imaging device using XIAO-ESP32-C6, OV5642 camera, and power-aware capture logic.
- Iterated through three PCB revisions and five 3D-printed enclosure revisions to improve reliability and manufacturability.
- Bench-tested capture consistency across lighting and viewing angles to validate robustness.

WatchFall – **Samsung Solve for Tomorrow** — Sept 2025 – present

- Built a fall-detection system with Raspberry Pi Pico, TensorFlow Lite inference, and buzzer alerting for real-time on-device monitoring.
- Engineered a hybrid dataset and sensor integration strategy to lower false positives and improve detection responsiveness.

Leadership

Chief Technical Advisor / Programming Lead, ITKAN Robotics – Plano, TX — July 2025 – present

- Directed software strategy for a multi-team robotics program with ~20 programmers, standardizing code structure, GitHub workflows, and integration practices.
- Implemented reliability engineering and contingency planning that preserved system performance after critical sensor failures.

Education

Wakeland High School, High School Diploma – Frisco, TX — Aug 2023 – May 2027

- Completed PLTW engineering coursework in digital electronics, systems design, and engineering development.
- Took rigorous course load, including Multivariable Calculus and Physics C

Boston University, Research in Science & Engineering (BU RISE) – Boston, MA — June 2026 – Aug 2026

- Researched under Dr. Wenchao Li, Department of Electrical and Computer Engineering
- Studied VLM/VLA-driven Sawyer robots to enable long-horizon task completion from vague human prompts